

The Grain Value Loop

An integrated rice biorefinery model that turns the byproducts India's government already procures — and currently gives away — into edible oil, food-grade derivatives, energy, and exports. Feedstock the state owns; demand the state can contract.

IMPORT SUBSTITUTION

₹21,000 cr

per year, across crops

BRAN GIVEN AWAY
TODAY
₹7,650 cr

in free CMR byproduct

CAPITAL SUBSIDY

35–50%
PMFME · PMKSY ·
PLISFPI

FCI-warehouse-MSPvsMSV · figures verified against DFPD, DGCI&S, MoFPI, real project DPRs

THE PARADOX

India imports what it grows — and gifts the feedstock

The government procures ~550 LMT of rice and ~300 LMT of wheat a year, then hands the milling byproducts to private millers for free and imports the finished derivative.

WHAT THE STATE DOES TODAY

- Procures paddy at MSP, hands it to CMR millers

WHAT THE STATE IMPORTS

- 57% of edible oil (~₹1.68 lakh cr/yr) — while #1 in rice-bran oil

- Millers keep ~180 LMT husk + ~65 LMT bran — the ₹7,650 cr byproduct giveaway
- Sells surplus grain raw (OMSS) at ~2–4% trader margin, often at a loss

- 67 LMT of pulses, vital wheat gluten, specialty starch, Vitamin-E, silica
- The *derivative* — while holding the raw material

World's #1 rice-bran-oil producer, yet imports more than half its cooking oil. The same pattern repeats for gluten, starch, protein, and nutraceuticals across every crop the state procures.

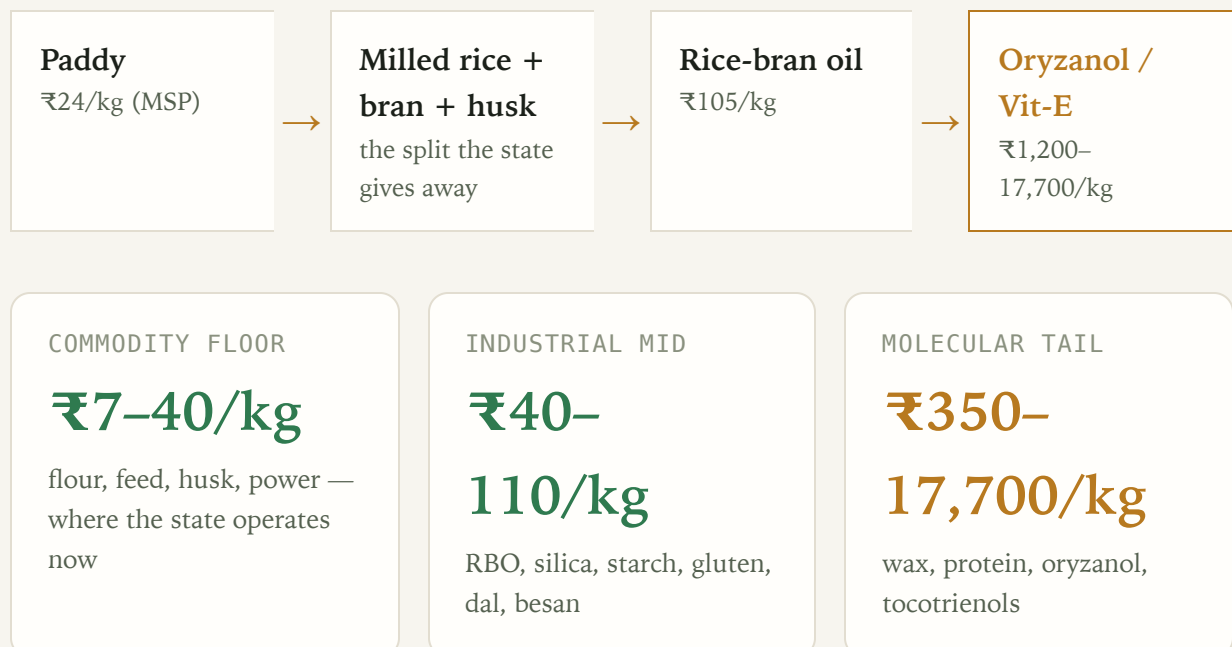
Sources: DFPD Annual Report 2024–25 • DGCIS FY26 • NITI Aayog edible-oil

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THE INSIGHT

Value density rises ~1,000× from grain to derivative

The margin is in the cascade, not the commodity — the sugarcane model (sugar + ethanol + power + CBG) ported to grain.



THE PROJECT

An integrated rice biorefinery — one complex, many streams

A cooperative/PPP-hosted complex, sited in an operational Mega Food Park, that takes government CMR paddy and runs the full cascade instead of milling and selling raw:

- Rice mill Bran stabilisation
- RB0 solvent extraction RB0 refinery
- Oryzanol / wax recovery
- Husk-fired power Husk-ash → silica
- Broken rice → poha/murmura
- Fortified rice kernels DORB → feed

WHY INTEGRATED

- **Husk powers the process** (already standard mill practice) — energy self-sufficient
- **DORB feed credit** makes the oil cheap (₹60 vs ₹105/kg)
- One feedstock stream → 8+ revenue lines
- The distillery/mill is the natural aggregation host

Edible oil first (import substitution at ₹105/kg), fuel only from the non-edible residual — India must diverge from the US "distillers-oil-to-biodiesel" model.

Grounded in NCDC cooperative rice-mill DPR + SEA (30 MT existing crush capacity)

RAW MATERIAL · PRICES & SOURCES

Feedstock the state already owns

INPUT	INDICATIVE PRICE	SOURCE / BASIS

Paddy (common)	₹2,369/qtℓ	MSP KMS 2025-26 · govt CMR procurement (~832 LMT)
Rice bran	₹2,000/qtℓ (₹20/kg)	NCDC DPR · to oil extractors; 16–22% oil
Rice husk	₹2,000/MT	NCDC DPR · fuel-grade; ~20% of paddy
Broken rice	₹1,500/qtℓ	NCDC DPR · poha/starch/ethanol feedstock
De-oiled bran (DORB)	₹15/kg (credit)	feed byproduct — the credit that makes RBO viable

AGGREGATION MODEL

PACS / FPO / cooperative societies — *already the paddy-procurement centres* — aggregate paddy + bran. The feedstock chain exists; it is currently dispersed to 100,000+ private mills.

SCALE AVAILABLE TO GOVERNMENT

~832 LMT CMR paddy → ~40 LMT government-touchable bran (41% of national) → ~6.4 LMT RBO at full capture.

Prices: NCDC "Establishment of Rice Mill" DPR (Chhattisgarh) · DFPD MSP · vintage noted

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UNIT ECONOMICS · FROM REAL DPRS

Margin and IRR — ranked, government-DPR-backed

PRODUCT	NET MARGIN	IRR	DPR SOURCE
Poha / murmura	10–15%	<3y payback	NIFTEM-T PMFME ✓
RBO solvent extraction	8–15%	–	IMARC/EIRI
Rice mill (anchor)	8.5–15%	19%	NCDC ✓
Maize wet-milling (starch)	12–20%	22–28%	industry
Dal mill	~5.3%	25–31%	RACP + NIFTEM ✓
Roller flour mill	3–5.5%	20–29%	RACP ✓

Margin $\neq$ return. The staples show thin 3–5% margins but **19–31% IRR** — low capex, fast turnover. The "process, don't sell raw" case rests on return-on-capital, not margin-on-sales.

Sources: NCDC · Rajasthan RACP (Grant Thornton) · NIFTEM–Thanjavur PMFME · IMARC – verified/reported labelled

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SCHEMES · THE FUNDING STACK

35–50% of capital comes from MoFPI

PMFME

35%

credit-linked capital
subsidy, max ₹10 L/unit ·
micro units (RBO, dal,
poha, besan)

PMKSY

35–50%

grants for clusters, cold
chain, agro-processing
(50% NE/difficult) · **the**
integrated complex

PLISFPI

6-yr

incentive on incremental
sales + branding-abroad ·
large branded (RBO
brand, exports)

MEGA FOOD PARK INFRASTRUCTURE

23 operational parks (of 41), ~₹120 cr
each with ~₹50 cr PMKSY grant —
ready common facilities (power, cold
chain, warehousing, effluent) to plug
into, not build.

THE SUBSIDY IS A MARGIN MULTIPLIER

Anchor DPR: ₹80 L cost → ₹16 L
subsidy + ₹52 L loan + ₹12 L own. The
19% project IRR becomes a much higher
return on the ₹12 L actually invested.
Capital-heavy units (RBO, silica, wet-
milling) gain the most from a 35–50%
grant.

Sources: pmfme.mofpi.gov.in · PMKSY FAQs · mofpi.gov.in/PLISFPI · MoFPI 41-MFP list (31.01.2023)

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Demand the government can contract

CAPTIVE · SUREST

- Fortified rice kernels — ₹17,082 cr program to 2028, 100% GoI-funded; ~3.5–4 LMT/yr standing demand
- Fortified atta · Vitamin-E fortificant from own bran

DOMESTIC GROWTH

- RBO — India #1 market, \$1.39 bn; + exports \$104.5 mn
- Poha/murmura (snacks 14.5% CAGR) · besan
- Maize starch \$4.1 bn · wheat gluten

EXPORT-LED

- Basmati ₹50,000 cr (FY25, +15.7%, ₹82/kg)
- Plant protein (14% CAGR) · spices ₹39,994 cr · green silica (Tata ₹775 cr)

The offtake is **contractable, not consumer-dependent**: food-processing supply agreements + fortification mandates + institutional (PDS/ICDS) offtake form a demand backbone independent of the retail shelf — the ethanol-blending-mandate lever, applied to processed grain.

Sources: APEDA FY25 (basmati ₹50k cr) · DFPD FRK program · DGCI&S · industry market reports – verified/reported labelled

THE PRIZE · IMPORT SUBSTITUTION

₹21,000 cr/yr across the crops the state holds

STREAM	IMPORT ₹/T (DGCI&S)	SAVINGS ₹ CR/YR	OFFTAKE
Rice bran oil	97,970	7,544	frying/HoReCa + fortification
Pulses (dal margin)	50,888–74,163	~7,000	dal millers + PDS
Cottonseed oil	98,766	1,975	food processing
Maize starch	37,170	1,115	food/pharma/paper

Husk → silica	86,730	867 tyre/industrial
Protein · gluten · Vit-E · lecithin	115k–17.7m	~1,900 processors + export

HIGH-CONFIDENCE
TIERS

₹18–21k cr

oil + pulses + industrial
mid

ON FEEDSTOCK THE
STATE

already

owns

via CMR procurement

THROUGH OFFTAKE IT
CAN

contract

not hope for

Source: consolidated-savings-by-crop · DGCI&S FY26 import unit values · illustrative, additive

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A CONCRETE MSV CASE · WHEAT → PASTA

Durum wheat: the fastest-growing value-add India under-grows

The same "import the derivative, hold the feedstock" pattern — in wheat. India is **wheat-surplus** and sells it cheap via OMSS (~₹2,550/qtl), yet:

- Pasta market **\$1.21 bn, growing ~14–16%/yr** — the fastest wheat-product segment
- India is the **world's #2 durum-semolina importer** — pasta needs durum (hard) wheat, which India barely grows (MP's Malwa belt, ~1.5–2 MT)
- DGCI&S: pasta imports **+24%**, biscuit imports **+28%**, gluten **+35%** — the value-added tier's imports are building

THE DOUBLE PLAY (DURUM MSV)

- **Procure durum by variety** — a variety-specific MSV that de-risks durum cultivation (the wheat parallel to the oilseed-MSP production lever)
- **Mill semolina → pasta** on OMSS/durum wheat, in a Mega Food Park
- Substitutes **both** the pasta imports AND the durum-wheat imports

- **Export prize:** India ships only \$63 mn of pasta into a global import trade of US \$1.62 bn • Germany \$1.24 bn — under 4% of the US market, on cheap OMSS wheat Italy can't match

- + vital gluten (13,000 t/yr imported) from the same wheat wet-mill

India is a net *exporter* of biscuits (Parle-G, 180 countries) — so the target is the **premium/pasta/durum tier** where imports are growing, not the mass market. Import substitution is already visible: durum-pasta imports fell –25% YoY.

Sources: DGCI&S TradeStat EIDB (exact 8-digit HS, FY25–26 prov.) • industry market sizes • report/wheat-value-added-products.md

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A CONCRETE CASE • EDIBLE OIL FROM FCI/CMR STOCK REUTILISATION

The oil India imports for ₹1.68 lakh cr — from grain the state already holds

India imports 168 LMT of edible oil — US\$19.3 bn, ₹1.68 lakh cr a year (57% of demand), landing at ~₹100/kg crude (palm ₹96). Yet the 24 MSP crops are the domestic oil-feedstock base — and FCI's own procured paddy throws off the cheapest oil in the country as a byproduct:

- **Rice bran oil** — bran from the ~130 MT of paddy under CMR; ₹60/kg ex-mill vs ₹105 import parity = ₹45/kg edge. India is the world's #1 RBO producer — and hands the bran to millers free (the ₹7,650 cr giveaway)
- RBO is also the *better* oil: γ-oryzanol, tocotrienols, phytosterols; cholesterol-

IMPORT-SUBSTITUTION SAVINGS ESTIMATE

STREAM	₹/KG EDGE	₹ CR/YR
Rice bran oil (7.7 LMT)	45	7,544
Cottonseed oil (2 LMT)	~10	1,975
Soy/sun crush — cake + forex*	system	~1,500
Capturable now		~₹11,000 cr/yr

*at-parity oils save via the oil-cake co-product imports don't carry + forex retained, not a retail-price cut.

lowering; high-heat stable; non-GMO (ICRBO-2016)

- **Cottonseed oil** — byproduct of cotton (an MSP crop): crush margin + cake

Honest tiering: ~₹11,000 cr/yr is **capturable now** from byproduct oils on feedstock the state already procures. The **production lever** (NMEO-OS oilseeds 39→69.7 MT) is the ceiling — closing even half the 168-LMT gap replaces ~₹84,000 cr of imports over 5–7 yrs — but mustard/groundnut are premium oils, a *volume/self-sufficiency* lever, never an in-year cheaper-cooking-oil promise.

Sources: DGCIS TradeStat EIDB (168 LMT/₹1.68 L cr) · DFPD edible-oil scenario · SEA (16.3 MT/₹1.61 L cr) · consolidated-savings-by-crop · report/msp-crops-refined-oil-source.md

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THE PDS · ONORC CASE · THE DEMAND RAIL ALREADY EXISTS

**₹2 lakh cr already spent, 80 cr served —
redirect the oil spend to domestic jobs**

DOMESTIC JOBS

FAIR PRICE SHOPS

5.43 lakh

99.8% ePoS-automated ·
dealer + staff livelihoods,
every district

RICE MILLS

75k–100k

10–50 workers each · TN
3,399 · AP 1,913 · WB
1,873

SOLVENT-EXTRACTION
PLANTS

PRODUCTION VOLUMES

Rice procured ~520
LMT

Wheat
procured 300
LMT

Paddy
milled/yr ~850
LMT

Rice bran oil ~10
LMT

Domestic
edible oil 122
LMT

Edible oil
imported 168
LMT

STATE EXPENDITURE

CENTRAL FOOD
SUBSIDY FY25

₹2.05 L cr

free rice/wheat to ~80 cr
(NFSA+PMGKAY) +
₹7,075 cr FPS margins

TN SPECIAL-PDS
OIL+DAL

₹3,800 cr

palmolein at ₹25/L on
~₹65/L subsidy — on an

~350

875 SEA members • 30 MT
processing capacity

Oilseeds 40.99
grown MT

The milling that makes
free PDS rice already
throws off the bran for
domestic oil.

import (AP/KA/TG/WB
run their own)

ONORC COVERAGE

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States/UTs • 20.54 cr cards
• portable, deduplicated

The state already spends ₹2 lakh cr+ and runs 5.43 lakh shops to feed 80 cr people — the demand rail is built. ONORC turns it into one portable, deduplicated national offtake: redirect the state oil subsidy from imported palmolein to domestic RBO, and the same rupee funds rice-mill, solvent-plant and oilseed-farm jobs instead of foreign crushers — import substitution and employment on infrastructure that already exists.

Sources: PRS Demand-for-Grants 2024-25 (Food & PD) • DFPD/PIB Year-End 2024 • TNCSC
• DCMSME • SEA of India • data/pds_onorc_jobs_volumes_spend.csv

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PER-STATE • WHO ALREADY SUBSIDISES OIL & PULSES IN THE RATION

The oil subsidy is real, state-run, and spent on imports

STATE	OIL IN PDS	PULSE IN PDS	CARD PRICE	SPEND ₹CR/YR
Tamil Nadu	Palmolein 1 L/card	Tur dal 1 kg	oil ₹25/L • dal ₹30/kg	3,800
Karnataka	Palm oil + Indira kit	Tur/moong 1 kg	dal ~₹38/kg	~360*
Gujarat	Edible oil + sugar	Chana + tur dal	subsidised	n/p
Andhra Pradesh	—	Tur dal 500 g	subsidised	n/p
Maharashtra	—	Chana + urad	₹35/kg	n/p
Telangana	—	Red gram dal	subsidised	n/p
Kerala (Supplyco)	Oil (below-mkt)	Pulses	ltd subsidy	n/p

TN PALMOLEIN BOUGHT

156 lakh L

+ 20,000 MT tur dal/yr —
all imported oil

STATES RUNNING OIL-PDS

4 major

TN · Karnataka · Gujarat ·
Kerala

STATES TAKING NAFED PULSES

11+

pulses far more
widespread than oil

Tamil Nadu is the flagship: ₹3,800 cr a year, almost entirely on *imported* palmolein — the single largest identifiable state oil subsidy a domestic-RBO switch would redirect. *Karnataka figure is dal-only (oil delivered in-kit); **n/p** = scheme confirmed on state portals but per-commodity spend not published — not zero. †NAFED chana, free to states, 2020.

Sources: TNCSC PDS scale · state civil-supplies portals · NAFED via Business Standard · press 2024-25 · data/pds_state_oil_pulse_expenditure.csv

THE FISCAL CASE · SAME RUPEE, WITH VALUE CAPTURE

₹2 lakh cr to give grain away — plus
₹2 lakh cr to import what it could make

WHAT THE SYSTEM COSTS NOW		WHAT PROCESSING RECOVERS	
MSP paid to ~1.3 cr farmers	₹2.32 L cr	Import substitution (processing)	₹21,087 cr
Food subsidy (free grain, 80 cr)	₹2.05 L cr	— of which capturable now	~₹11,000 cr
Edible-oil imports	₹1.68 L cr	Byproduct monetised (bran → RBO)	₹7,650 cr
Pulse imports	₹0.31 L cr	Excess-stock carry avoided	up to ₹18,000 cr
Excess-stock carry (~330 LMT over buffer)	₹18,000 cr		

Rice bran given away free ₹7,650 cr

No double-count: procurement sits *inside* the subsidy. The two clean burdens are the ₹2.05 L cr subsidy + ~₹2.0 L cr imports.

Processing-margin 3–5× uplift

Branded staple processing earns 11–12% EBITDA vs 2–4% raw trading — the money is in the processing, not the sourcing.

Nothing in the food-security guarantee is cut. Farmers still get MSP; 80 cr still get free grain. What changes: the byproducts stop being given away and the oil/pulses stop being imported — the *surplus and the byproduct are processed* instead of dumped and re-imported. A value-capture layer bolted onto a ₹4 lakh-cr/yr machine, not a new subsidy.

Sources: PIB/FCI procurement (wheat ₹62,156 cr RMS25–26) · PRS food subsidy ₹2.05 L cr · DGCIS imports · FCI stock 738 LMT (1 May 25) · report/fiscal-rationalisation-procurement-vs-processing.md

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NATIONAL BUILD-OUT · STORAGE × FARMER-AGGREGATION

The grain and the farmers are in different states — design for it

STATE	FCI STORAGE LMT	SFAC FPOS
Madhya Pradesh	200	348
Punjab	175	78
Haryana	100	119
Uttar Pradesh	47	758
Bihar	21	312
Andhra Pradesh	25	247
Rajasthan	16	225
Maharashtra	24	213

FCI+hired storage (Nov-24 bulletin, ~841 LMT all-India) · SFAC 10,000-FPO registry (3,649, Feb-25 —

THE MISMATCH = THE DESIGN RULE

- **Storage belt** (Punjab 175, MP 200, Haryana 100) holds the grain — but is thin on FPOs (Punjab 78)
- **FPO belt** (UP 758, Bihar 312, AP 247, Rajasthan 225) has the farmers — but far less storage
- **MP is the one state with both** — 200 LMT + 348 FPOs + soybean/pulses/mustard = **the natural first hub**

Siting logic: oilseed/pulse crush → FPO-dense deficit states (MP, Maharashtra, Rajasthan, AP); rice-bran → the paddy/storage belt (Punjab, Haryana, Chhattisgarh). Match each stream to the

Sources: DFPD Foodgrain Bulletin Nov-2024 (storage) · SFAC state-wise 10,000-FPO list 24-02-2025 · data/national_fci_fpo_by_state.csv

THE VEHICLE · DON'T CREATE A PSU – REVIVE THE ONE DFPD OWNS

DFPD already owns a dormant edible-oils processing PSU

Per igod.gov.in, DFPD's org chart already holds **three** delivery bodies:

FCI

procurement · storage · the grain + bran feedstock

CWC

the warehousing network

HVOC — Hindustan Vegetable Oils Corp

a CPSE built for vegetable-oil / vanaspati processing — now dormant

India needn't *create* a food-processing PSU — it can *revive* HVOC: the corporate shell, the DFPD mandate and the edible-oil remit already exist, alongside the grain (FCI) and the storage (CWC).

BUILD IT LIKE THE SURVIVORS, NOT THE FAILURES

- **Infrastructure + processing, NOT trading** — STC→PEC→NCEL died trading naked
- **FPO-fed supply** — the AMUL/NAFED survival trait
- **Ring-fenced, contracted offtake** — PDS/ONORC oil + fortification, not the open market
- **JV/PPP capital** — HVOC brings mandate + sites; partner brings crush/refining
- **Sit on the FCI/CWC network** + the ₹7,650 cr free bran

A revived HVOC is the **edible-oil counterpart to FCI**: FCI holds the grain, HVOC processes the oil, CWC stores, FPOs supply, ONORC distributes — anchoring the ₹1.68 lakh-cr import substitution and supplying the PDS oil ration domestically instead of buying imported palmolein.

THE MODEL · WHY GOVERNMENT-COOPERATIVE-PPP

Ring-fence the commercial arm; contract the offtake

STRUCTURE

- **PACS/FPO** aggregates paddy + bran (already the procurement centre)
- **Private partner** builds + runs RBO extraction, silica, power (the capital + know-how)
- **State** provides VGF/PMKSY grant + a fortified-rice / RBO **offtake guarantee**
- Ring-fenced from MSP/buffer policy — the COFCO lesson, not the STC/Thailand failure

WHY NOT LEAVE IT TO PRIVATE MILLERS

- 100,000 mills do basic CMR — but won't invest in bran stabilisation, RBO, silica, cogen
- The integrated **value-added** mill is the PPP white space (verified: largely untried)
- The offtake guarantee is the revenue backbone — mirrors PEG's silo-hire guarantee

History's warning (verified): STC→PEC→NCEL failed by trading naked; CWC, AMUL, Kendriya Bhandar succeeded by infrastructure / farmer-ownership / ring-fencing. This is an *infrastructure + processing* play, not a trading arm.

Sources: PPP rice-mills · PSU-agri-ventures-history · agri-trader-playbook

One integrated complex, in an operational food park

ANCHOR: RICE MILL

₹80 L

1 TPH, 19% IRR (NCDC
DPR)

+ RBO EXTRACTION

₹15–40 cr

8–15% margin, the ₹7,544
cr anchor

+ POHA/MURMURA LINE

₹21 L

10–15% margin, <3y
payback

+ HUSK POWER/SILICA

cluster

energy self-sufficiency +
tyre-grade silica

CAPITAL SUBSIDY

35–50%

PMKSY cluster + PMFME
units

OFFTAKE SECURED BY

contract

FRK/PDS + food
processors + fortification

The proof-points already exist as verified DPRs and operational parks. The novelty is **integration + the offtake guarantee** — assembling proven units into one government-anchored complex that captures the cascade instead of gifting it.

All component economics from real project DPRs; integration is the un-built step

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WHY IT MATTERS · DOUBLING FARMERS' INCOME

Real farm income is ~a third of the way to doubling

The 2016 goal: double farmers' *real* income by 2022-23. Deflated by CPI-AL (the farm household's own cost basket, MoSPI), the **Real Farm Income Index** reads:

2016-17
RFII 99



2021-22
RFII 131



Target

THE LEVER THE GAP NEEDS

- Crop income is only **37%** of farm income — price alone can't double it
- The other 63%: wages, livestock, **non-farm/value-addition**
- Cutting *cost* defends real income against inflation the nominal

200 = doubling

Achieved ~5.7%/yr real vs the 10.4%/yr needed — reaching 200 slips to ~2030.

headline hides

- Exactly the byproduct-cascade + processing-margin this project builds

RFII is a monitorable metric: CPI-AL deflator refreshes monthly (MoSPI), income on each SAS/NAFIS survey.

Sources: NABARD NAFIS 2021-22 · NSSO SAS · MoSPI CPIALRL · Dalwai Committee (DFI)

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THE FARM-SIDE LOOP · FUEL CHOICE

Diesel→CNG + residue: the inflation-proof income lever

FUEL SHIFT
(DIESEL→CNG)

+ ₹5,400

/yr per 1.5-ha paddy household · 24% on tractor fuel · +3.3 RFII pts

FULL CBG LOOP
(+RESIDUE)

+ ₹9,750

/yr · residue→CBG + FOM slurry · +5.9 RFII pts · closes 9% of the doubling gap

MILL FUEL CHOICE

**₹1 vs
₹10.45**

husk/CBG per kWh vs grid · ₹2.7-7.9 cr/yr on the mill's EBITDA

The loop closes: crop → residue → CBG → fuels the tractor (24% cheaper, inflation-proof) AND the mill's boiler → slurry to the field. Diesel is a volatile imported input; CBG from the farmer's own residue defends the *real* income the doubling metric measures.

CACP A2+FL · PIB CNG-tractor (PRID 1697555) · SATAT CBG · TN diesel/CNG Jul 2026

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Stop gifting the byproduct. Contract the demand. Capture the cascade.

India is the world's #1 producer of the raw byproduct and a net importer of the refined derivative — for oil, gluten, starch, protein, and Vitamin-E alike. The feedstock is procured, the subsidy is 35–50%, the parks are built, the DPRs are proven. What's missing is the integrated complex and the offtake guarantee. That is the project.

SAVINGS

₹21k cr/yr

BEST DPR MARGIN

**poha 10–
15%**

BEST DPR IRR

dal 25–31%

Full analysis, data & sources: github.com/herrrickshaw/FCI-warehouse-MSPvsMSV ·
research synthesis, not investment advice